Note

Generally, $f \circ g \neq g \circ f$ for any two functions f and g. So, composition of functions is not commutative.

Example 1.20 Represent the function $f(x)=\sqrt{2x^2-5x+3}$ as a composition of two functions.

Solution $f_2(x)=2x^2-5x+3$ and $f_1(x)=\sqrt{x}$

$$f(x) = \sqrt{2x^2-5x+3} = \sqrt{f_2(x)}$$
$$= f_1\left[f_2(x)\right] \qquad = f_1 f_2(x)$$

Example 1.21 If $f(x)=3x-2$, $g(x)=2x+k$ and if $f \circ g = g \circ f$, then find the value of k.

Solution $f(x)=3x-2$, $g(x)=2x+k$

$$f \circ g(x) = f(g(x)) = f(2x+k) = 3(2x+k)-2 = 6x+3k-2$$

$$f \circ g(x) = 6x+3k-2\,.$$

$$g \circ f(x) = g(3x-2) = 2(3x-2)+k$$

$$g \circ f(x) = 6x-4+k\,.$$

Given that $f \circ g = g \circ f$

$$\therefore 6x+3k-2 = 6x-4+k$$

$$6x-6x+3k-k = -4+2 \Rightarrow k = -1$$

Do You Know?

The Composition $g \circ f(x)$ exists only when range of f is a subset of domain of g.

Example 1.22 Find k if $f \circ f(k) = 5$ where $f(k) = 2k-1$.

Solution $f \circ f(k) = f(f(k))$

$$= 2(2k-1)-1 = 4k-3\,.$$

$$f \circ f(k) = 4k-3$$

But, $f \circ f(k) = 5$

$$\therefore 4k-3 = 5 \Rightarrow k = 2\,.$$

1.9.1 Composition of three functions

Let A, B, C, D be four sets and let $f : A \to B$, $g : B \to C$ and $h : C \to D$ be three functions (Fig.1.43). Using composite functions $f \circ g$ and $g \circ h$, we get two new functions like $(f \circ g) \circ h$ and $f \circ (g \circ h)$.

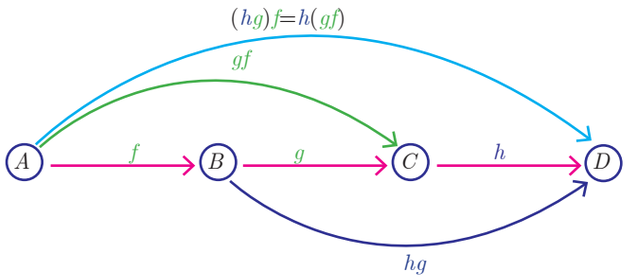


Fig. 1.43

We observed that the composition of functions is not commutative. The natural question is about the associativity of the operation.

Note

Composition of three functions is always associative. That is, $f \circ (g \circ h) = (f \circ g) \circ h$

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